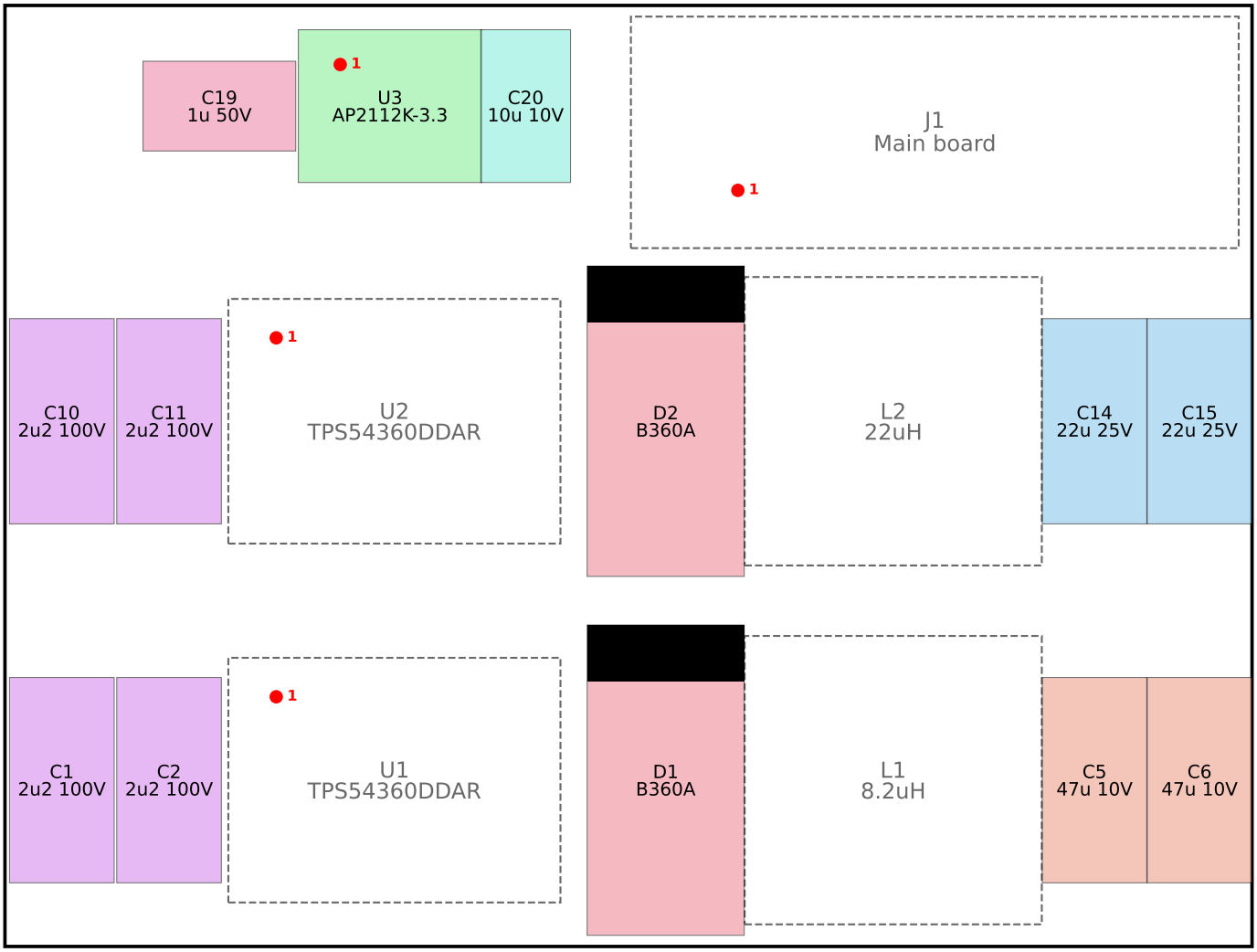
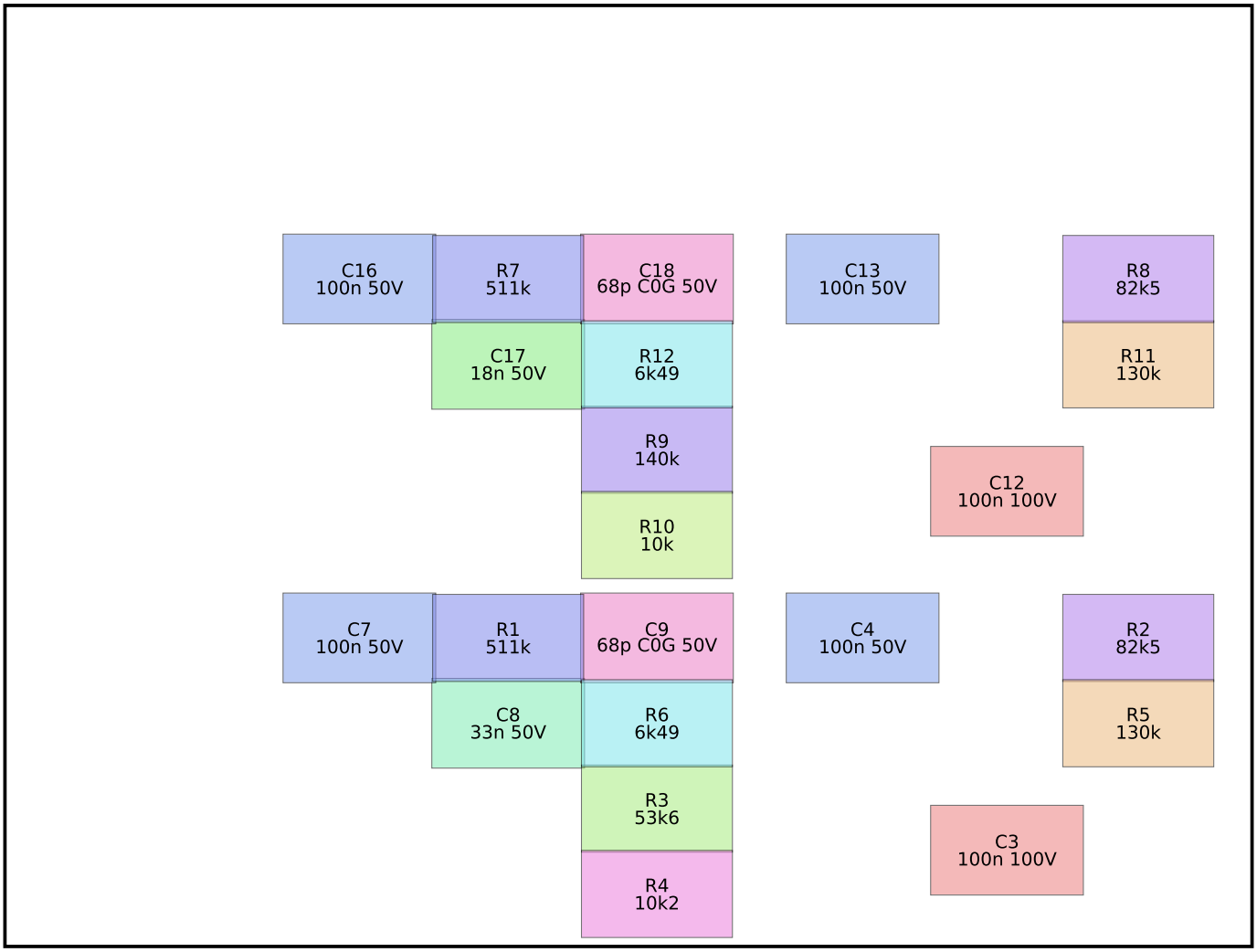
power - TOP

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



power - BOTTOM (seen from below, mirrored)

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



power - values

- 100n 100V x2 bottom: C3, C12
- 100n 50V x4 bottom: C4, C7, C13, C16
- 10k x1 bottom: R10
- 10k2 x1 bottom: R4
- 10u 10V x1 top: C20
- 130k x2 bottom: R5, R11
- 140k x1 bottom: R9
- 18n 50V x1 bottom: C17
- 1u 50V x1 top: C19
- 22u 25V x2 top: C14, C15
- 22uH x1 top: L2
- 2u2 100V x4 top: C1, C2, C10, C11
- 33n 50V x1 bottom: C8
- 47u 10V x2 top: C5, C6
- 511k x2 bottom: R1, R7
- 53k6 x1 bottom: R3
- 68p C0G 50V x2 bottom: C9, C18
- 6k49 x2 bottom: R6, R12
- 8.2uH x1 top: L1
- 82k5 x2 bottom: R2, R8
- AP2112K-3.3 x1 top: U3
- B360A x2 top: D1, D2
- Main board x1 top: J1
- TPS54360DDAR x2 top: U1, U2

power - key

Marks on the board pages

red dot + 1 : pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules

black bar   : diode cathode, matching the band printed on the diode body

red +   : + pad of electrolytic / tantalum capacitors

no mark   : resistors, ceramic capacitors, inductors, fuses, crystals, pin strips, bidirectional TVS

orange corner: part fitted by hand (not placed by JLCPCB); dashed outline: large part, fit it last

the BOTTOM page is mirrored: it shows the board as you see it from below

Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pin 1 is left of it on the two-leg side

SOT-23-5 / SC-70 / SOIC / VSSOP / QFN / LGA: a dot, dimple or bevelled edge marks pin 1

Addressable LEDs: a marked (cut / green) corner - check the datasheet, it is easy to misread

Diodes: the band is the cathode (K)

Electrolytic capacitors: the stripe marks the - side (the other pad is +)

Tools

0805 / SOT-23: a 3-5x magnifier or a 10x loupe

QFN / LGA / DFN / SC-70: a USB microscope (check the pads under the part after reflow)